

CINECA AGENTIC PLATFORM

An Enterprise-Grade Multi-Tenant AI Orchestrator with MCP Tools and Secure Graph Querying

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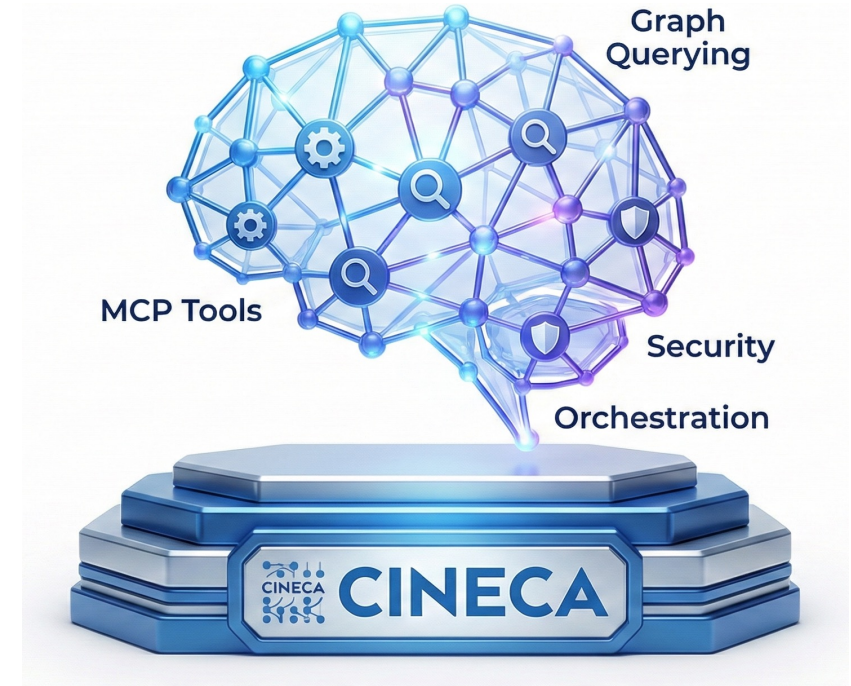
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CINECA

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PRODUCTION GAP

The gap between what users need and what current tools provide.

The Problem

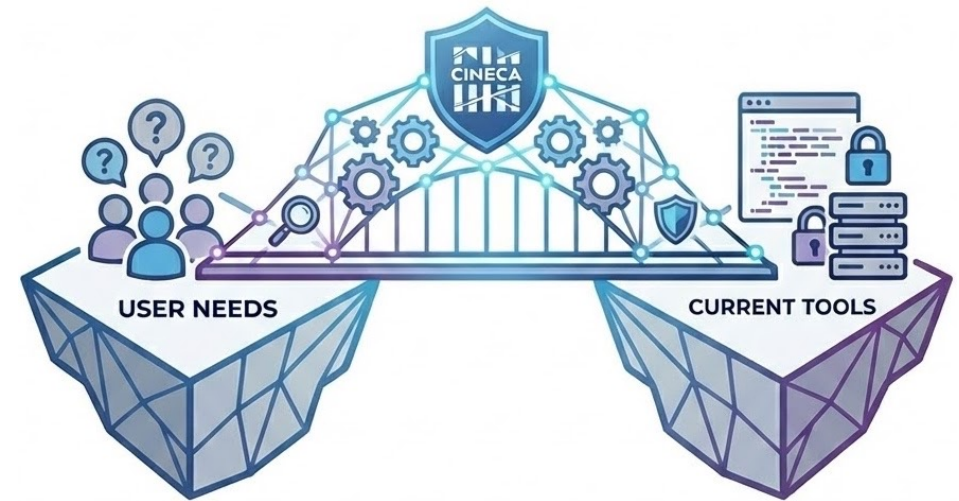
- Most users cannot write Cypher, so they depend on developers to access graph data. This creates delays and keeps data locked behind technical complexity.

Why Simple Chatbots Are Not Enough

- Basic chatbots can answer questions, but they cannot
 - handle multi-step tasks
 - query databases or run jobs
 - enforce security and access control
 - provide audit trails and traceability

Why This Matters for CINECA

- CINECA needs a platform that lets users query graph data directly in natural language, while remaining:
 - Secure
 - Scalable
 - Reproducible
 - cost-efficient
 - flexible across LLM providers

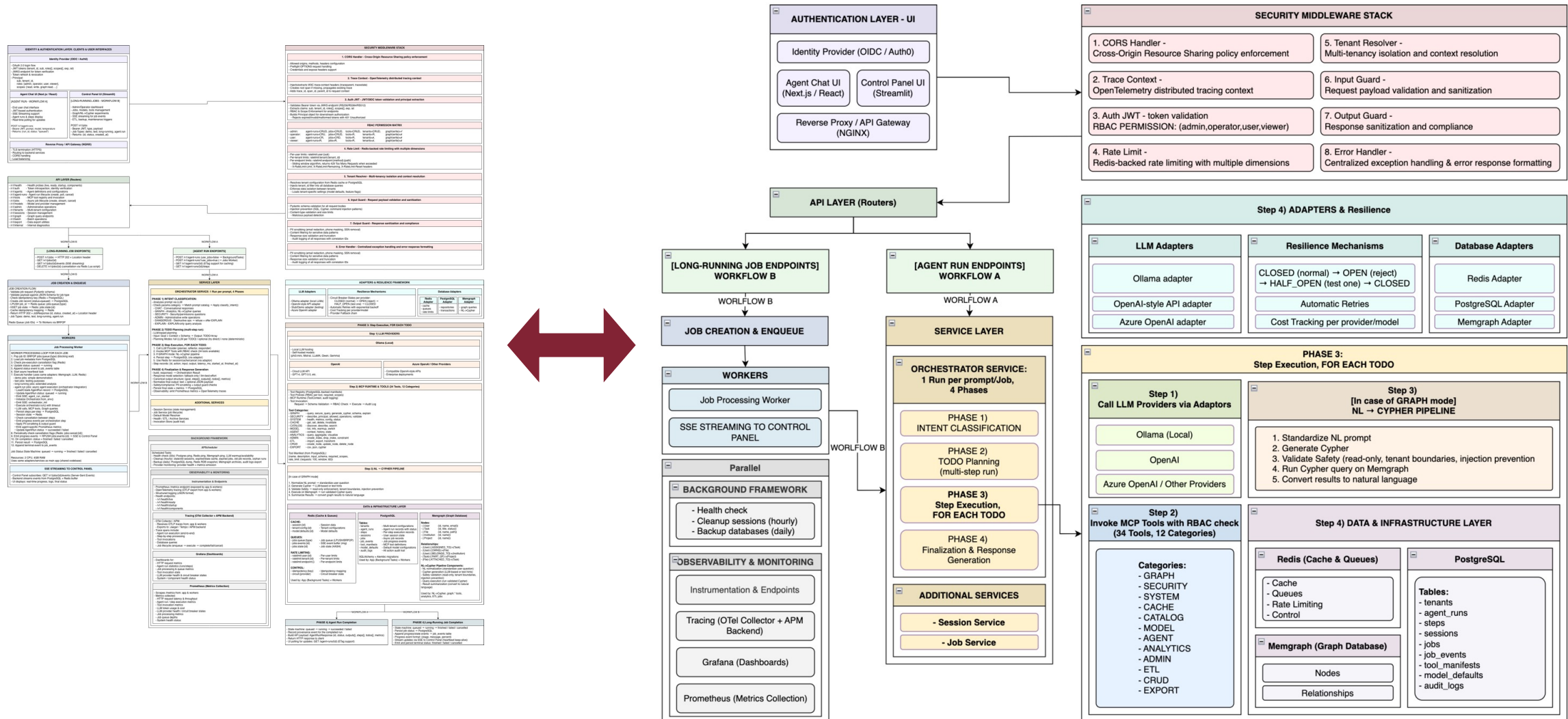


Simplified



FINAL ARCHITECTURE

Full



WHAT WAS DELIVERED

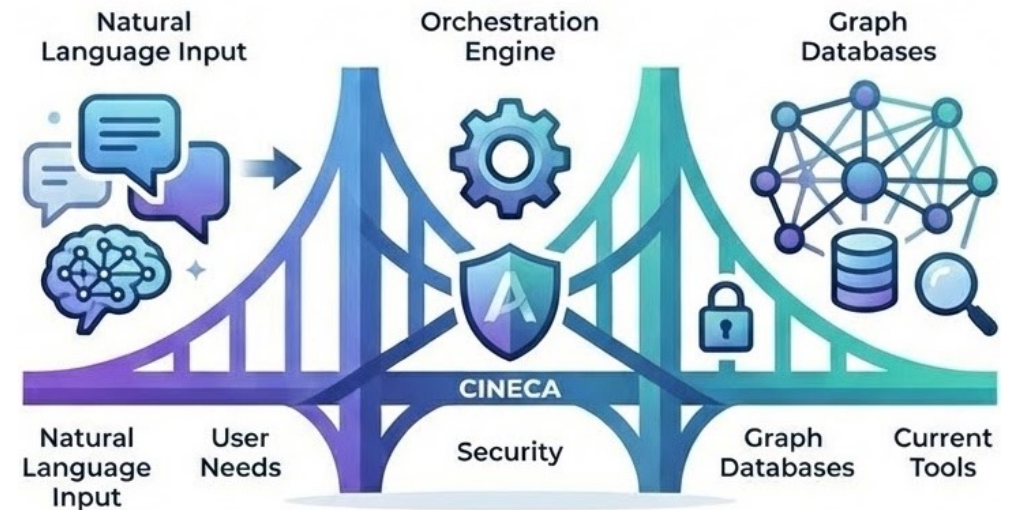
A production-ready agentic platform for secure natural-language access to graph data.

Core Delivery

- End-to-end platform: UI, APIs, orchestration, workers, and a three-database backend
- Safe 4-phase agent workflow with NL→Cypher capabilities
- Multi-tenant security and support for multiple LLM providers
- Reliability, background processing, and observability built in

Production Evidence

- 16+ platform components (databases, queues, gateway, tracing)
- 76 API endpoints across 16 router groups (e.g. auth, tools, jobs, graph)
- 34 tools across 17 categories with governed execution
- 3,000+ automated tests (unit, integration, security, end-to-end)
- ~411,700 total lines of code



AUTHENTICATION LAYER & UI

Users sign in through an identity provider, receive a JWT, and access the platform through a secure gateway.

Authentication & Access

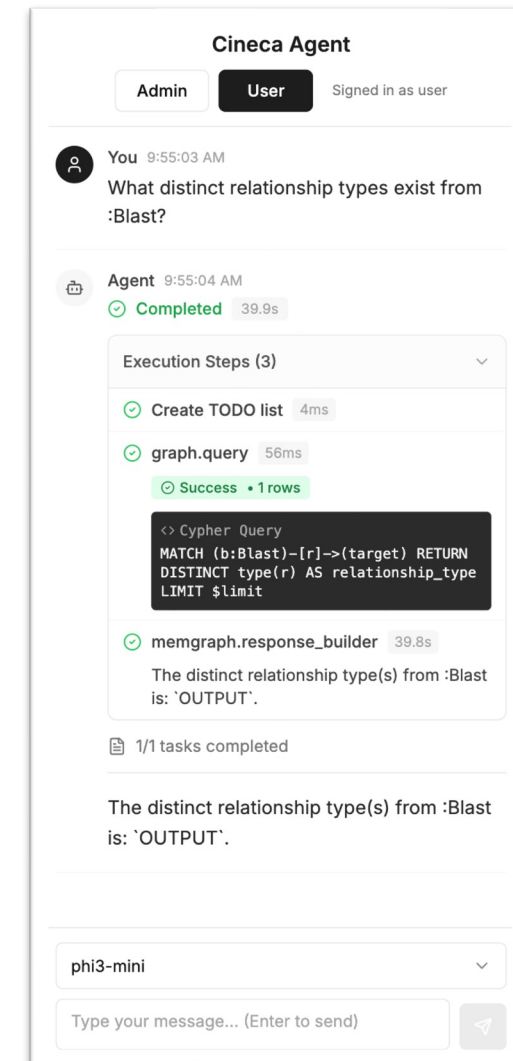
- Users log in with OAuth 2.0 and receive a JWT with their identity, tenant, role, and permissions
- Every request verifies the token and applies role-based access control
- The UI adapts to different access levels

Agent Chat UI

- Main interface for asking questions in natural language and interacting with agents
- Sends authenticated requests to the backend
- Shows live progress, execution steps, final results, and the selected model

Gateway

- NGINX handles secure access, routing, and CORS



SECURITY MIDDLEWARE STACK (Part 1/2)

Every request passes through 8 Security middleware layers before reaching the business logic.

1. CORS Handler

Controls which websites can call the API.

- Example: requests from the official chat UI are allowed; unknown domains are blocked.

2. Trace Context

Assigns a trace ID so one request can be followed across the whole system.

- Example: one user request can be tracked through the API, orchestrator, and database.

3. JWT Auth + RBAC

Checks the user token and enforces role-based permissions.

- Example: a viewer tries to delete a run and gets 403 Forbidden.

4. Rate Limiter

Prevents abuse by limiting how many requests can be sent.

- Example: after too many requests in one minute, the user gets 429 Too Many Requests.

Role	agent-runs	jobs	tools	tenants	Graph (write)
admin	CRUD	CRUD	CRUD	CRUD	✓
operator	CRU	CRUD	R	R	×
user	CR	CRD	R	×	×
viewer	R	R	R	×	×

SECURITY MIDDLEWARE STACK (Part 2/2)

Every request passes through 8 Security middleware layers before reaching the business logic.

5. Tenant Resolver

Ensures each user only accesses data from their own organization.

- Example: a user from one tenant cannot see data from another tenant.

6. Input Guard

Validates requests and blocks unsafe or malformed input.

- Example: an injection-like prompt is rejected before execution.

7. Output Guard

Cleans responses before they are returned to the user.

- Example: an email address in the output is redacted.

8. Error Handler

Returns safe, standardized errors without exposing internal details.

- Example: a database failure returns a generic error plus a correlation ID.



API LAYER - Routers

The platform exposes 76 endpoints grouped by domain under versioned APIs.

Core API Groups

- /v1/health → service and dependency health checks
- /v1/auth → authentication and token validation
- /v1/agents → agent definitions and configuration
- /v1/tenants → tenant settings, limits, and feature control
- /v1/models → LLM provider and model management
- /v1/tools → MCP tool discovery and invocation

Execution APIs

- /v1/jobs → long-running jobs, live progress, and cancellation
- /v1/agent-runs → start runs, track status, get results, cancel execution
- /v1/graph → direct Cypher queries and NL→Cypher execution
- /v1/sessions → session and conversation state management

Operational APIs

- /v1/admin → maintenance tasks, cache control, and system stats
- /v1/export → export runs, job results, and audit logs



JOB CREATION

Client submits a long-running task → the system validates it, avoids duplicates, and queues it safely.

1. Request Validation

- Checks required fields and payload structure
- Example: agent.run must include prompt, user_id, and tenant_id

2. Idempotency Check

- Looks for an existing job with the same idempotency key
- Example: if the client retries after a timeout, the same job is returned

3. Job Record Creation

- Creates a new PostgreSQL job record with status = queued
- Example: the job is stored before execution starts

Supported Job Types

- demo, test, long-running, agent.run

4. Queue Enqueue

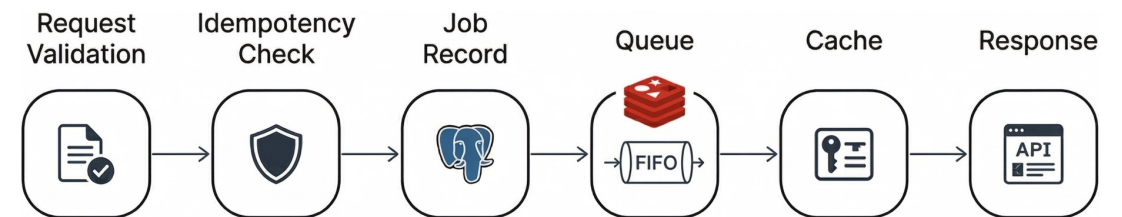
- Pushes the job ID into the Redis queue for workers
- Example: agent.run jobs go to jobs:queue:agent.run

5. Idempotency Cache

- Stores key → job ID mapping in Redis for fast duplicate detection
- Example: future retries can return the same job immediately

6. HTTP Response

- Returns 202 Accepted with the job ID and status
- Example: client receives a Location header to poll /v1/jobs/{id}



WORKERS + AGENT.RUN HANDLER → JOB PROCESSING ENGINE (Part 1/2)

Background workers pick up queued jobs and execute full agent workflows reliably outside the API process.

1. Acquire Job

- The worker waits for a job in Redis and takes exclusive ownership when one appears
- This prevents the same job from being processed twice

2. Load Job Data

- Using the job ID, the worker loads the full job record from PostgreSQL
- PostgreSQL is the source of truth; Redis only provides the queue entry

3. Check Early Cancellation

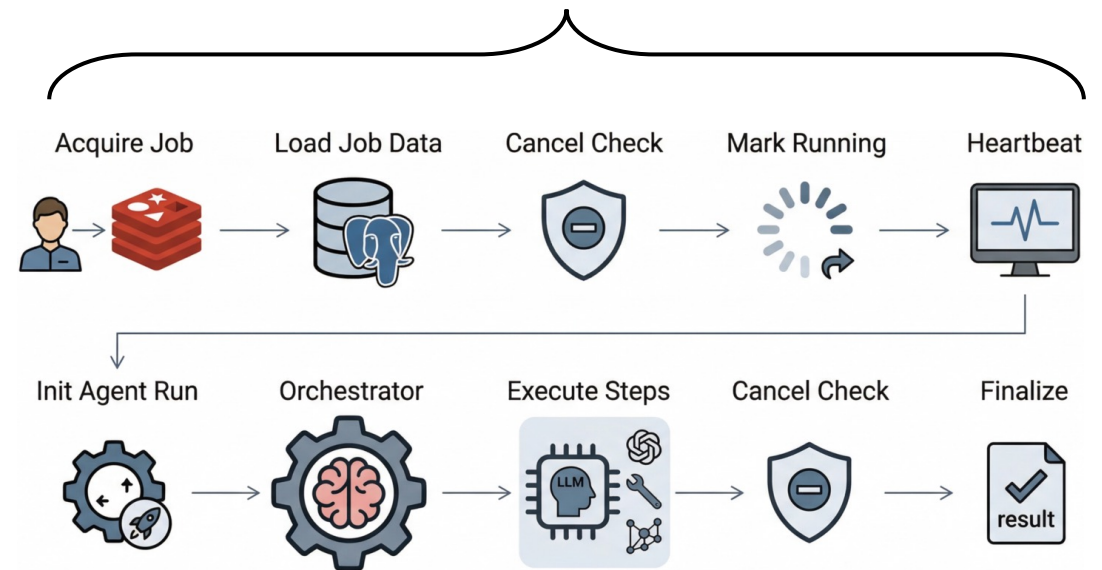
- Before starting, the worker checks whether the job was cancelled
- If already cancelled, execution is skipped safely

4. Mark as Running

- The worker updates the job state from queued to running
- This state change is stored for tracking, audit, and live updates

5. Start Heartbeat

- While the job is active, the worker sends periodic heartbeat updates
- If the heartbeat stops, the system can detect a stuck or failed worker



WORKERS + AGENT.RUN HANDLER → JOB PROCESSING ENGINE (Part 2/2)

Background workers pick up queued jobs and execute full agent workflows reliably outside the API process.

6. Initialize Agent Run

- For agent.run jobs, the worker creates an AgentRun record linked to the job
- Startup events are emitted so the UI can show immediate progress

7. Run the Orchestrator

- The worker calls the orchestrator as the core execution engine
- The orchestrator plans and executes the workflow step by step

8. Execute and Persist Steps

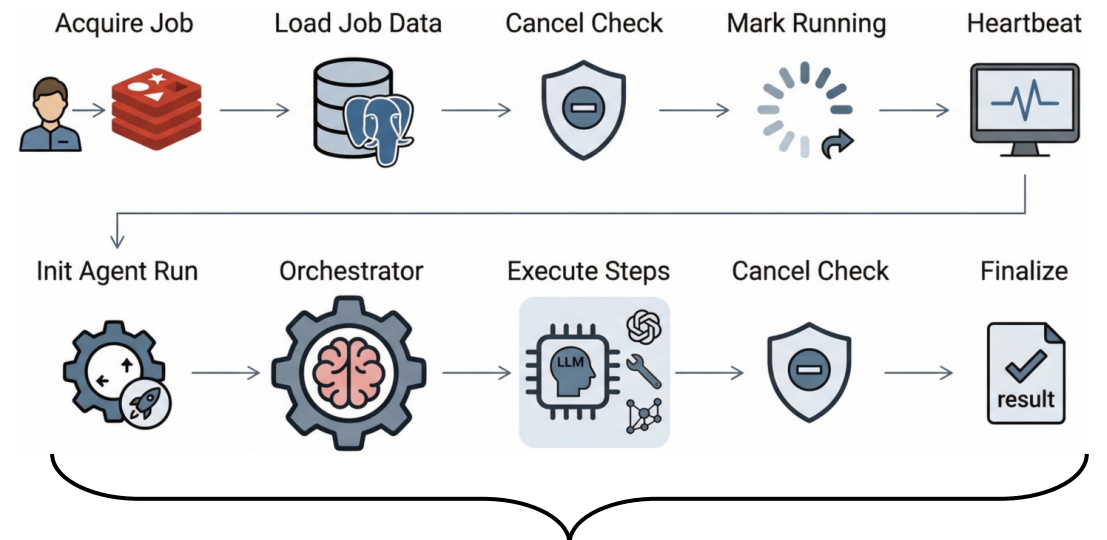
- Each step may call an LLM, invoke MCP tools, or query the graph
- Completed steps are saved immediately so progress is not lost

9. Check Cancellation During Execution

- Between steps, the worker checks again whether cancellation was requested
- If cancelled, it stops cleanly after the current completed step

10. Emit Progress and Finalize

- Progress events are streamed live, outputs are post-processed for safety, and metrics are recorded
- At the end, the final result is saved and the job is marked as succeeded or failed



SERVICE LAYER → ORCHESTRATOR SERVICE (Part 1/3)

The orchestrator contains the core business logic and processes each prompt in 4 phases.

Phase 1 → Intent Classification

- The orchestrator first determines what kind of request the user is making
- It checks a prompt catalog first; if no match is found, it uses an LLM classifier

Main intent types include:

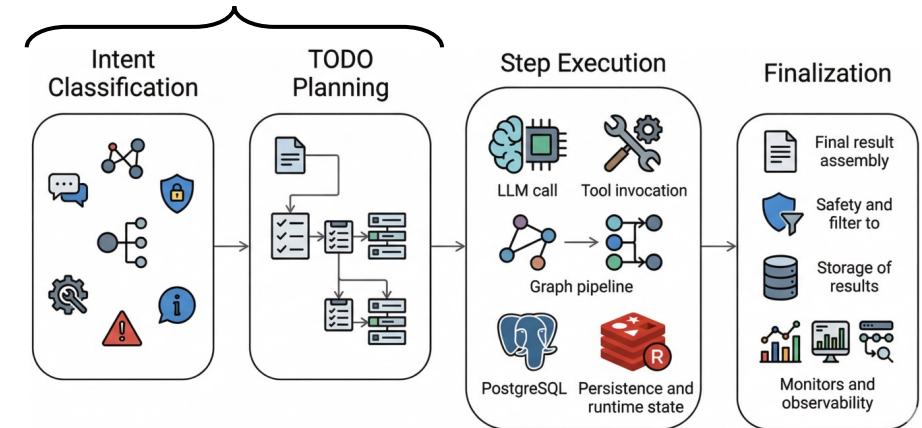
- CHAT → normal conversational response
- GRAPH → NL→Cypher query
- SECURITY → permission or access question
- ADMIN → administrative action
- DANGEROUS → destructive request, refused safely
- EXPLAIN → explain an action without executing it

Phase 2 → TODO Planning

- For multi-step tasks, the orchestrator turns the prompt into an ordered TODO plan
- Planning uses the user request, conversation context, and available schema/tool information
- The output is a structured sequence of actions, such as query → analyze → summarize

Planning Modes:

- full → creates a detailed step-by-step plan for complex tasks
- optional → tries direct execution first, plans only if needed
- none → uses a fixed path for predefined or catalog-matched requests



SERVICE LAYER → ORCHESTRATOR SERVICE (Part 2/3)

The orchestrator contains the core business logic and processes each prompt in 4 phases.

Phase 3 → Step Execution

The orchestrator executes the TODO plan step by step using LLMs, MCP tools, the graph pipeline, and runtime state.

1. Call LLM Providers

- Uses different LLM roles for planning, checking results, and generating responses
- Calls go through adapters with retries, fallbacks, and circuit breakers

2. Invoke MCP Tools

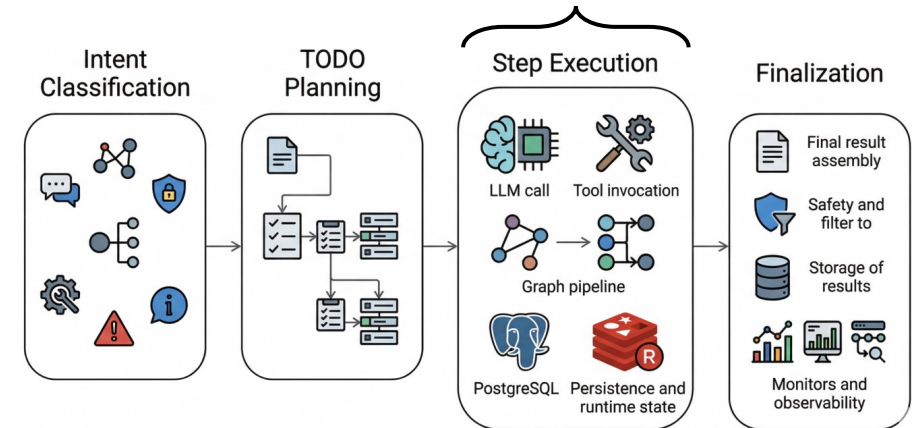
- Calls tools for graph queries, security checks, cache access, and export
- Each tool call is permission-checked and keeps session context

3. Run the Graph Pipeline

- For graph requests, the system runs a 6-stage NL→Cypher flow:
 - normalize input → check catalog match → generate Cypher → validate safety → execute on Memgraph → summarize results.
- Validation includes syntax, tenant boundaries, query depth, timeout, result limits, and read-only enforcement

4. Save Progress and Manage State

- Each step is saved to PostgreSQL for traceability and recovery
- Redis is used for caching and cancellation checks



"Show 5 Blast pairs outputting to same BlastedSeq"

```
1. Normalize → "blast pairs same blastedseq limit 5"
2. Catalog → No match → LLM
3. Generate →
MATCH (b1:Blast)-[:OUTPUT]->(s)<-[:OUTPUT]-(b2:Blast)
WHERE b1<>b2 LIMIT 5
4. Validate → ✓ Read-only ✓ Tenant ✓ Depth ✓ Timeout
5. Execute → Returns 5 pairs
6. Summarize →
"Found 5 Blast pairs sharing BlastedSeq targets..."
```

SERVICE LAYER → ORCHESTRATOR SERVICE (Part 3/3)

The orchestrator contains the core business logic and processes each prompt in 4 phases.

Phase 4 → Finalization and Response Generation

The orchestrator assembles the final result, applies safety checks, saves the outcome, and records monitoring data.

1. Build the Final Response

- Combines all completed step outputs into one final answer
- Can fall back to a cached or default response if needed

2. Normalize the Output

- Returns a consistent output format, including user-facing text and optional JSON

3. Apply Safety Checks

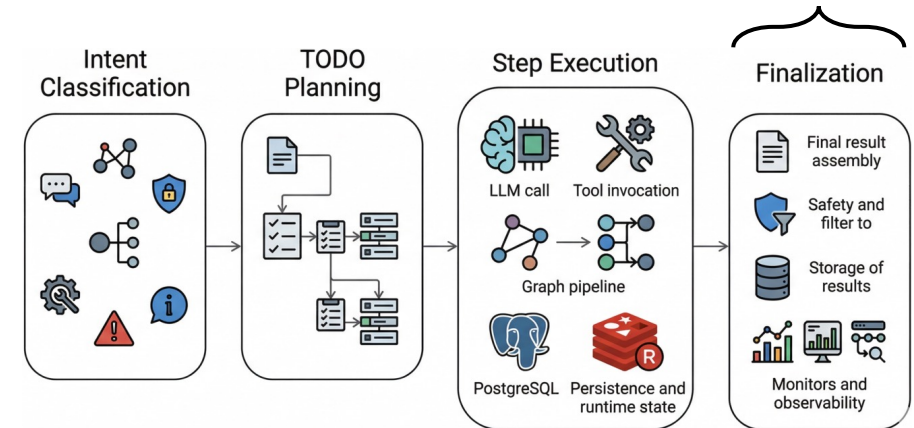
- Removes sensitive data and filters unsafe content before sending the response

4. Save the Final State

- Stores the final status, output, and metrics in PostgreSQL for audit and analysis

5. Emit Observability Data

- Records metrics and traces for dashboards, debugging, alerting, and monitoring



```
{  
  "goal": "Find collaborating institutions",  
  "steps": [step1, step2, step3],  
  "outputs": ["Found 5 institutions..."],  
  "todos": [completed_todo1, completed_todo2],  
  "metrics": { "duration_ms": 2340, "tokens": 1850 }  
}
```

LLM PROVIDERS → MODEL-AGNOSTIC ARCHITECTURE

The platform can use multiple LLM providers, both local and cloud, without changing the application logic.

Local Models → Ollama

- Supports running open models locally on your own infrastructure
- Useful for data control, lower cost, offline use, and custom setups
- Example models: Phi-3, Mistral, LLaMA, Qwen, Gemma

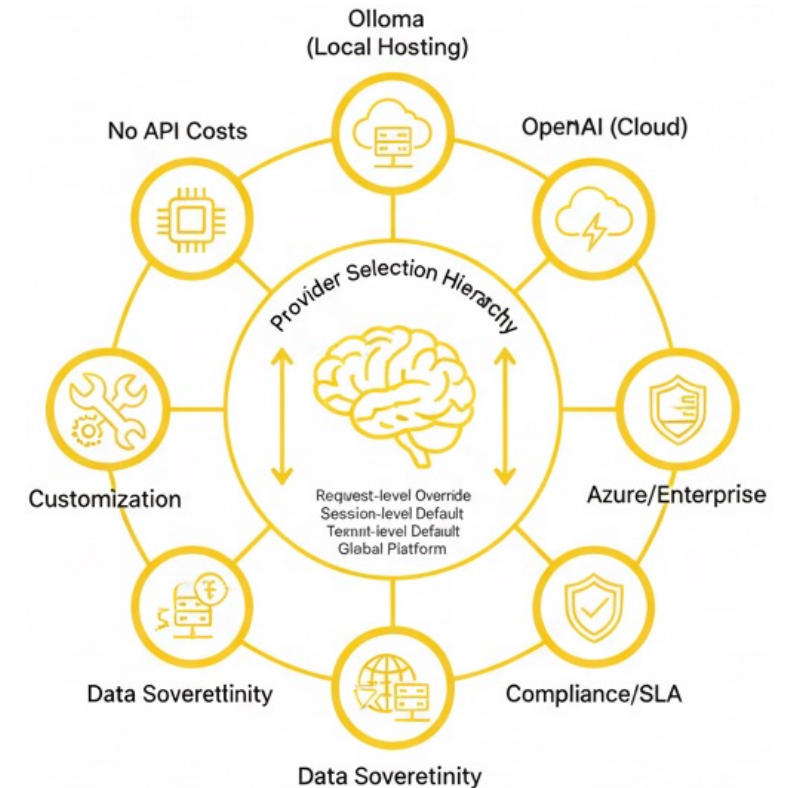
Cloud Models → OpenAI / Azure / Others

- Supports hosted providers for strong model quality and managed infrastructure
- Useful for easy deployment, enterprise compliance, and regional hosting
- Examples: OpenAI, Azure OpenAI, and compatible providers

Provider Selection

- A model can be chosen at different levels:
- This gives flexibility to users, teams, and platform administrators

Request-level override
↓ (if not set)
Session-level default
↓ (if not set)
Tenant-level default
↓ (if not set)
Global platform default



MCP RUNTIME & TOOLS - 34 Tools, 12 Categories

Every MCP tool the agent can use is registered, validated, permission-checked, and audited.

Core Ideas

- Tool definitions are stored in a registry with schema, required scopes, and limits
- Each tool call is checked against user permissions before execution

Tool Invocation Flow

Request → validate input → check permissions → execute tool → audit → return result

Main Categories

- Graph → query Memgraph, inspect schema, generate or explain Cypher
- Security → inspect identity, permissions, and allowed actions
- System → check platform health, metrics, configuration, and status
- Catalog → discover available resources, tools, and schemas
- Model → list, inspect, warm up, or switch LLMs
- Agent → access conversation context, history, and session state
- Analytics → run aggregations and prepare visualization data
- Admin → perform restricted database administration actions
- ETL / CRUD / Export → import data, update graph entities, and export results in formats such as CSV or JSON

Examples

- Graph tools can answer a natural-language question by generating and running Cypher
- Security tools can tell a user what actions they are allowed to perform
- Export tools can return query results as CSV or JSON



DATA LAYER → REDIS, POSTGRESQL, MEMGRAPH

The platform uses three databases, each with a different role: speed, durability, and graph relationships.

Redis → Real-Time Runtime State for speed and real-time coordination

It stores:

- cache data, sessions, tenant config, and model defaults
- rate limits for users, tenants, and endpoints
- job queues, job state, and progress events
- cancellation flags, idempotency keys, and circuit-breaker state

PostgreSQL → Durable Platform State for reliable and transactional system data

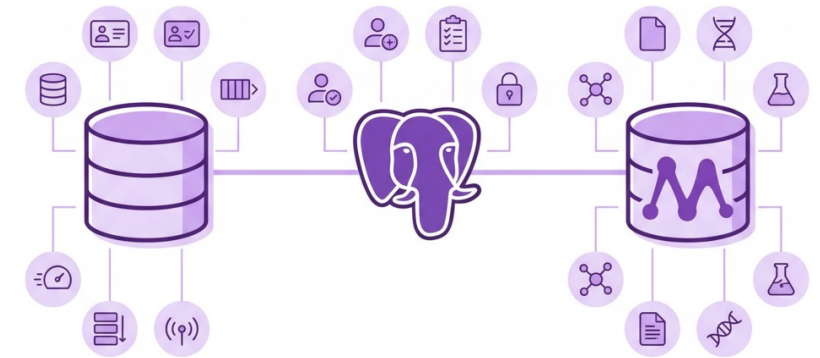
It stores:

- tenants, sessions, runs, steps, and jobs
- job events, tool definitions, and tool invocation history
- model defaults, user preferences, and provider configs
- encrypted secrets, audit logs, and idempotency records

Memgraph → Bioinformatics Graph State for connected bioinformatics data and efficient Cypher queries

It stores:

- graph nodes such as users, institutions, tasks, files, datasets, samples, experiments, publications, genes, proteins, pathways, tools, workflows, and results
- graph relationships such as WORKS_AT, RUNS, INPUT, and OUTPUT
- node and edge metadata such as roles, status, timestamps, tags, and file details



ADAPTERS + RESILIENCE

The platform stays flexible and reliable by using swappable adapters and automatic failure handling.

Adapters

- The platform uses swappable adapters for both LLM providers and databases
- This keeps the business logic independent from specific vendors or storage systems
- For LLMs, the same internal interface can work with Ollama, OpenAI, Azure OpenAI, or stub/demo providers
- For databases, the same logic can call cache, queue, relational, or graph operations through Redis, PostgreSQL, and Memgraph adapters

Why This Matters

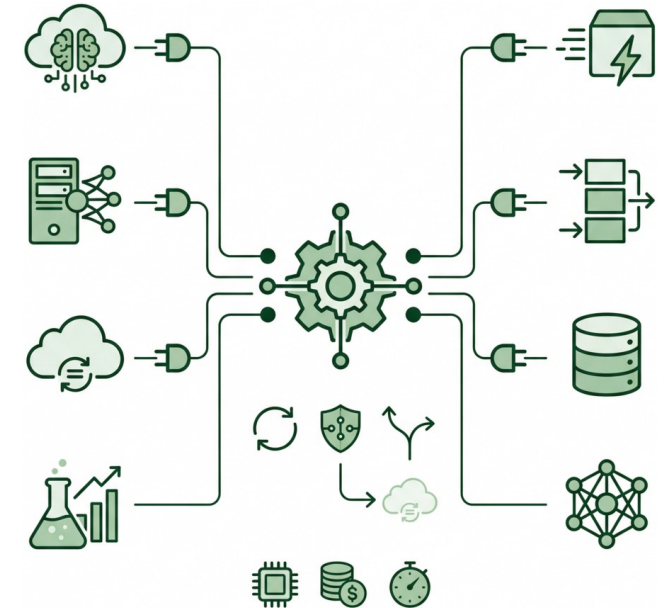
- Providers or backends can be changed without rewriting the orchestrator
- Example: switching from OpenAI to Azure only changes the adapter, not the rest of the code
- The same idea applies to database access such as `cache.get()` or `graph.query()`

Resilience

- The platform does not fail immediately when a provider has problems
- It uses retries with backoff, circuit breakers, and fallback providers
- If one provider becomes unhealthy, requests can fail fast or move to another provider
- Circuit breakers move between closed, open, and half-open states to control recovery

Operational Tracking

- Every LLM call is measured for tokens, cost, and latency
- This supports monitoring, budgeting, and capacity planning



BACKGROUND TASKS + OBSERVABILITY

The platform runs scheduled maintenance tasks and collects traces & metrics for visibility, reliability, & recovery.

Background Tasks

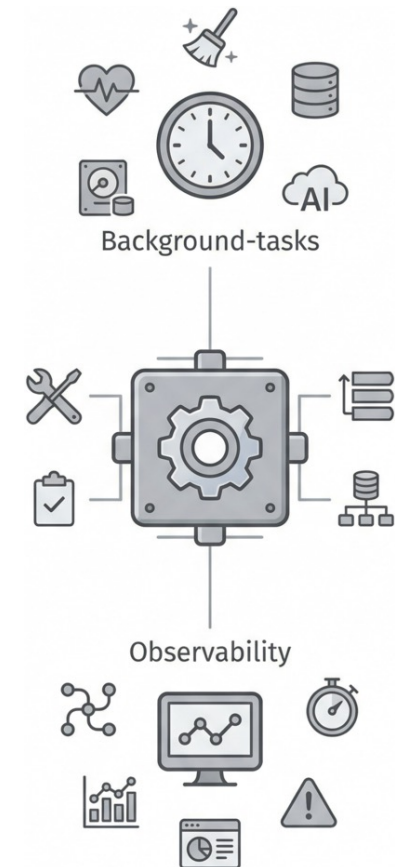
- APScheduler runs scheduled jobs automatically in the background.
- Health checks regularly verify PostgreSQL, Redis, Memgraph, and LLM providers.
- Cleanup jobs remove expired sessions, old cache entries, completed jobs, and orphaned runs.
- Daily backups protect PostgreSQL, Redis, Memgraph, and audit data.
- Provider monitoring updates health status, circuit breakers, and metrics.

Observability

- OpenTelemetry traces each request from end to end, including LLM calls, tool invocations, database queries, and job execution.
- Prometheus collects metrics from both the app and workers, including HTTP traffic, runs, steps, tools, jobs, LLM usage, circuit breakers, and system health.
- Grafana dashboards turn these traces and metrics into views of performance, failures, queue depth, provider health, costs, and overall system status.

Why This Matters

- Scheduled background tasks help keep the platform healthy, clean, and recoverable.
- Traces and metrics make it easier to diagnose slow requests, failing tools, unhealthy providers, and system-wide issues.



COMPARISON WITH STATE OF THE ART

Capability-by-capability comparison of your CINECA Agentic Platform against Top-10 similar platforms

Platform	Full Stack (UI + API + Jobs)	Orchestration Durability (Retries, Resume)	Agent Planning Loop (Multi-Step Tools)	Tool Ecosystem & Schema Controls	Security & Governance (RBAC, Audit)	Graph NL→Cypher Support (Secure + Tenant-Aware)	Observability (Tracing, Metrics, LLM Logs)	LLM-Agnostic (Model Provider Independent)	License / Pricing	Typical Limitation Compared to CAP	Fit Score
CINECA Agentic Platform	✓ Full-stack: UI + API + Jobs	✓ Durable: retries + checkpoints	✓ Built-in MCP loop engine	✓ Native registry + schema audit	✓ JWT, RBAC, tenancy, I/O guards	✓ Built-in NL→Cypher with validation & tenancy	✓ Full telemetry + LLM evals	✓ Yes (OpenAI, Ollama, etc.)	Custom-built (Internal only)	—	★ 5.0
Temporal	✗ Engine only (no UI/tools)	★ Best-in-class workflow durability	✗ No agentic planning loop	~ Pluginable tools (not MCP-native)	✓ Fine-grained RBAC + auth options	✗ No graph layer	✓ Metrics + traces	✓ Yes	Free OSS (MIT) + Paid Cloud	No agentic loop or UI; not a full platform	★★ 4.0
Argo Workflows	~ K8s-native stack	✓ DAGs + Retry Semantics	✗ No agent loop	~ Container steps only	~ Basic RBAC via K8s	✗ No Cypher / graph features	~ Limited metrics	✓ Yes	Free OSS (Apache-2.0)	K8s-focused; lacks planning & tool UX	★★ 3.5
LangGraph	✗ Library only	✓ Durable w/ state checkpointing	✓ Agent loop with state machine model	✓ Built-in agent tool patterns	✗ No RBAC / audit; DIY needed	~ Custom NL→Cypher possible	~ Partial (via adapters)	✓ Yes	Free OSS (MIT)	No orchestration stack; governance must be added manually	★★ 4.0
OpenAI Agents SDK	✗ SDK only	~ Runtime-level retry only	✓ SDK-defined agent planning	✓ Typed tools as functions	✗ No RBAC or tenancy	~ Custom Cypher interface possible	~ Logs via app code	~ Partial (OpenAI-focused)	Free OSS (MIT) + Paid OpenAI API	No job system, no multi-tenant security	★★ 3.5
Semantic Kernel	✗ Middleware only	~ Retry via plugins	✓ Plugin-based agent orchestration	✓ Tools + DI + planner interfaces	~ Basic auth via host app	~ Requires custom NL→Cypher logic	~ Custom logs via adapters	✓ Yes	Free OSS (MIT)	Middleware only; lacks orchestration/runtime infra	★★ 3.5
LlamaIndex	✗ RAG/agent SDK	~ Partial retries	✓ Tool-using planner + query agents	✓ Strong RAG support	✗ No RBAC, audit, tenancy	~ External Cypher logic possible	~ Logs via app code	✓ Yes	Free OSS (MIT) + Paid Cloud	No durability, security, or orchestration stack	★★ 3.0
Haystack	✗ Library only	~ Retryable components	✓ Agent + tools (non-durable)	✓ Tool abstraction + DSL	✗ No auth/governance	~ Cypher possible w/ custom nodes	~ Basic logs	✓ Yes	Free OSS (Apache-2.0)	Not production-grade orchestration	★★ 3.0
n8n	✓ GUI product + workflow UI	✓ Retry + error steps	~ Linear tool invocation (not agentic)	✓ Large integration catalog	~ Role-based app-level security	✗ Not graph-native	~ Workflow logs; no LLM eval	✓ Yes	Free OSS (SUL) + Paid Cloud SaaS	Good for automation; lacks agentic structure	★★ 2.5
Windmill	✓ UI + job scripting platform	✓ Durable jobs + cron	✗ No agent loop	✓ Script-based tool definitions	✓ RBAC, SSO, auditing	✗ Not graph-native	~ Run logs and dashboards	~ Partial (not LLM-centric)	Free OSS (AGPL mix) + Paid tiers	Good for internal workflows; lacks agent stack	★★ 3.5
Langfuse	✗ Observability layer only	✗ No execution	✗ No planning or tool use	~ Eval schema for prompts/tools	~ Some logging support	✗ Not applicable	✓ Best-in-class for LLM tracing	✓ Yes	Free OSS (MIT Core) + Paid SaaS	Tracing/monitoring only; no workflow or orchestration	★★ 3.0

COMPARISON WITH STATE OF THE ART

The CINECA Agentic Platform is more complete than typical libraries.

How CAP Differs:

- It provides a full stack, including UI, APIs, jobs, and orchestration
- It includes durable execution with retries, checkpoints, and background processing
- It supports an agent planning loop with MCP-based tool execution
- It adds strong security and governance through JWT, RBAC, tenancy, and output/input guards
- It includes built-in NL→Cypher support with validation and tenant isolation
- It also includes observability, LLM-provider flexibility, and production controls

How Other Platforms Compare:

- Temporal and Argo are strong for workflow durability, but they are not agent platforms
- LangGraph, OpenAI Agents SDK, Semantic Kernel, LlamaIndex, and Haystack support agent logic, but usually require extra work for security, durability, and platform operations
- n8n and Windmill provide useful workflow UIs, but they are not designed as secure agent orchestration platforms
- Langfuse is excellent for observability, but it does not execute workflows or agents

Main Takeaway

CAP stands out because it combines full-stack delivery, durable orchestration, governed tool execution, secure graph querying, and observability in one platform, while most alternatives are strong only in some of these areas

CONCLUSION + FUTURE WORK

Closed the production gap.

What Was Achieved

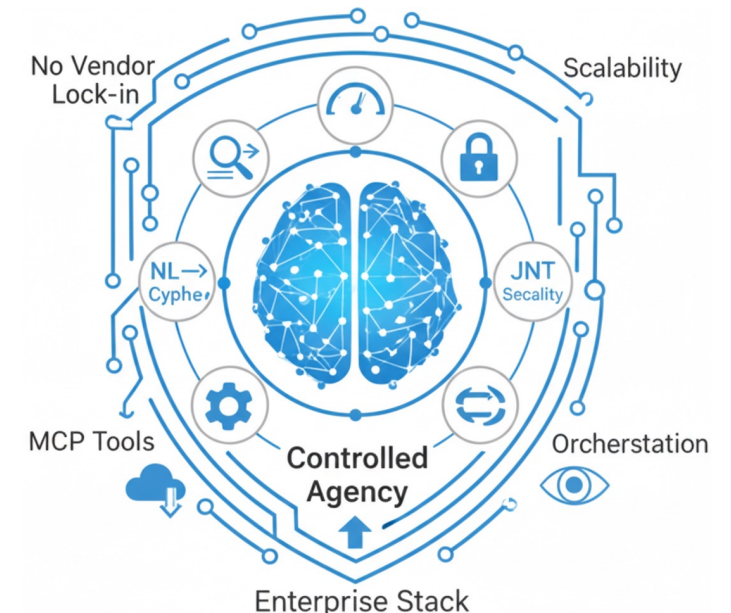
- The platform closes the production gap by turning natural language into safe, traceable graph actions
- It delivers a full enterprise stack with secure UIs, APIs, orchestration, workers, and a multi-database backend
- Agent runs are controlled, not just conversational, with MCP tools, RBAC, tenant isolation, and safe NL→Cypher execution
- It is production-ready, with large API coverage, governed tools, resilience mechanisms, and built-in observability
- It remains flexible across providers such as OpenAI, Azure, and Ollama

Key Contribution

- A secure and extensible NL-to-action platform ready to grow across tools, models, and tenants

Future Work

- Add an "Ask Mode" for lightweight question answering without full agent planning
- Make the orchestrator handle more prompt types more reliably, from simple to ambiguous or multi-step requests
- Improve recovery and reproducibility through stronger checkpointing and deterministic re-runs
- Add smarter scheduling with priorities, quotas, and backpressure under heavy load



Thanks For Your Attention

Feel free to ask your questions